



Smart Medicine Reminder System for DVAGO



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Problem Statement

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“A Missed Dose Is a Missed Chance”

Missing a dose can seriously affect treatment, especially for patients with long-term conditions. But most people don't forget on purpose; they just don't have a reliable system to remind them when it's time to take their medicine or refill their prescription.

Objective

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1. Schedule Medicine Reminders

Set dosage, frequency, and duration in just a few taps; minimal effort required.

2. Smart Notifications

Receive push notifications and voice alerts exactly when a dose is due.

3. AI-Assisted Medicine Scanning

Photograph the medicine packaging and let OCR auto-fill the details for you.

4. Adherence Tracking & Analytics

View your dose history and adherence trends across weekly, monthly, and yearly reports.

FYP Scope

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□ Included:

- Designing and developing a reminder system for the DVAGO platform.
- Notification and voice-based alerts for timely medicine intake.
- History tracking and user-configurable reminder settings.
- AI-based medicine recognition via image scanning for easier reminder setup.

□ Out of Scope:

- Live integration, backend access, or commercial deployment without DVAGO's approval.

Our Methodology

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The Smart Medicine Reminder System was built using the **Agile methodology**, following the **Scrum framework**.

Why Agile?

- ❑ Multiple features to build (scheduling, AI scanning, notifications, UI), needed **modular development**.
- ❑ Requirements can change during testing when needed **flexibility**.
- ❑ Regular supervisor feedback had to be incorporated quickly and needed **continuous improvement**.
- ❑ FYP-I and FYP-II had fixed deadlines and needed **steady, trackable progress**.

Our Project Plan

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Task	JUL	AUG	SEP	OCT	NOV	DEC	JAN	FEB	MAR	APR	MAY	Jun
Requirement Gathering												
UI Design & AI Setup												
Backend & Notification Setup												
Voice Alerts & Testing												
Final Report & Presentation												

Budget / Costing

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- ❑ Developers (2 @ Rs. 700/day x 4 days x 4 weeks = Rs.11,200/month x 12 = **PKR 134,400 est.**)
- ❑ Laptop (Rs. 150,000 x 2 = **PKR 300,000 est.**) — Already owned
- ❑ Hard Drive / Storage (Rs. 14,000 x 2 = **PKR 28,000 est.**) — Already owned
- ❑ Electricity (Rs. 5,000/month x 12 = **PKR 60,000 est.**)
- ❑ Internet (Rs. 3,000/month x 12 = **PKR 36,000 est.**)
- ❑ Firebase (Auth + FCM) — Spark Plan = **PKR 0** (Free tier)
- ❑ MongoDB Atlas — M0 Cluster = **PKR 0** (Free tier)
- ❑ Space OCR API — **PKR 0** (Free tier)
- ❑ Domain & Hosting = **PKR 0** (Free dev environment)
- ❑ Tools — VS Code, Postman, GitHub, Figma = **PKR 0** (Free/open-source)
- ❑ Miscellaneous = **PKR 10,000 est.**

Total Estimated Cost: PKR 568,400 est.

FYP Deliverables

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- ❑ **Working Android Mobile Application**
- ❑ **Software Requirements Specification (SRS) — v2.0**
- ❑ **Software Design Specification (SDS) — v2.0**
- ❑ **Test Plan & Test Case Report**
- ❑ **Final Year Project Report**
- ❑ **Fortnightly Sign-off Sheet**
- ❑ **Project Poster**
- ❑ **GitHub Links for All Artifacts**

Literature Review

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WHO reports medication adherence averages only 50% globally, dropping further in developing regions like Pakistan [3][4]. Mobile health apps have proven effective; JMIR studies confirm reminders improve adherence by over 20% [1][2]. Apps like Medisafe [5] and MyTherapy [6] demonstrate this in Western markets, yet intelligent reminder features remain largely absent from Pakistan's e-pharmacy platforms, a gap this project directly addresses.

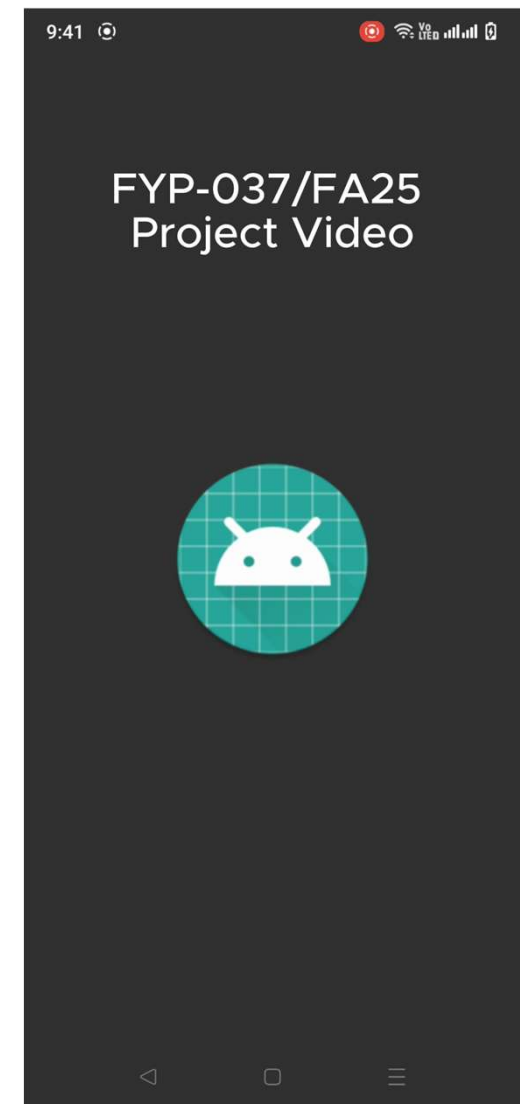
Demo of 100% of Work



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Key App Screens

- ❑ **Splash / Onboarding / Login / Sign-Up**
- ❑ **Dashboard**
(Next Dose + Live Countdown + Today's Schedule + Mark as Taken)
- ❑ **My Medicines (Search + Filter)**
- ❑ **Add Medicine**
- ❑ **AI Scan Medicine**
- ❑ **Analytics Dashboard**
- ❑ **Profile + Notification Settings**
(Push/Sound/Vibration/Voice)



Test Plan & Test Cases

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Test Plan — Smart Medicine Reminder System for DVAGO

The following table presents the detailed test plan, including the schedule for each functional component of the application.

S.No	Description	Tested by	Start Date	End Date
1	Login and Sign-Up	Abdul Ghani	06-Jun-2026	06-Jun-2026
2	Onboarding & Notification Permission	Abdul Ghani	07-Jun-2026	07-Jun-2026
3	Dashboard – Next Dose & Mark as Taken	Abdul Ghani	08-Jun-2026	08-Jun-2026
4	Add Medicine (Manual Entry)	Abdul Ghani	10-Jun-2026	10-Jun-2026
5	Scan Medicine (AI Wrapper Scan)	Abdul Ghani	10-Jun-2026	10-Jun-2026
6	My Medicines – List, Search & Filters	Abdul Ghani	11-Jun-2026	11-Jun-2026
7	Dose Reminder Notification (Lock Screen)	Abdul Ghani	11-Jun-2026	11-Jun-2026
8	Notifications Panel & Activity Log	Abdul Ghani	11-Jun-2026	11-Jun-2026
9	Analytics Dashboard (Adherence Tracking)	Abdul Ghani	11-Jun-2026	11-Jun-2026
10	Profile Management & Logout	Abdul Ghani	11-Jun-2026	11-Jun-2026

Reference

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- [1] "Improving Medication Adherence with Mobile Apps," *Journal of Medical Internet Research (JMIR)*, 2022.
- [2] A. Morrison et al., "Mobile Health Applications and Patient Self-Management," *Journal of Medical Internet Research (JMIR)*, 2021.
- [3] World Health Organization, *Adherence to Long-term Therapies: Evidence for Action*, 2019.
- [4] World Health Organization, *Report on Digital Health in Developing Countries*.
- [5] Medisafe, "Medication Reminder & Pill Tracker," [Online]. Available: <https://www.medisafe.com>. [Accessed: 15 May 2026].
- [6] MyTherapy, "Medication Reminder & Health Tracker," [Online]. Available: <https://www.mytherapyapp.com>. [Accessed: 15 May 2026].

Thank You!